DP Biology – B2.3

# Cell specialization

- What do stem cells do in multicellular organisms?
- How do differentiated cells suit their specialized roles?

SL statements B2.3.1–6. Textbook pp. 256–264.
Image: a sperm meeting an egg. Public domain.

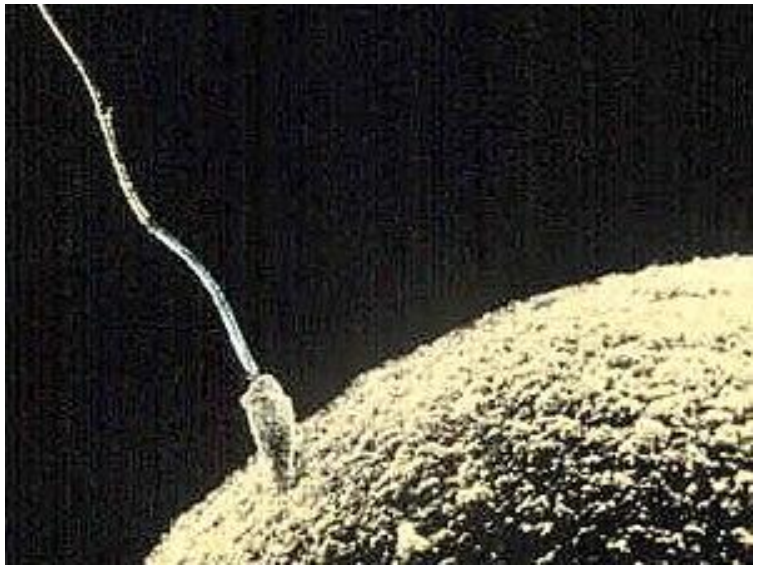


# Objectives for each class

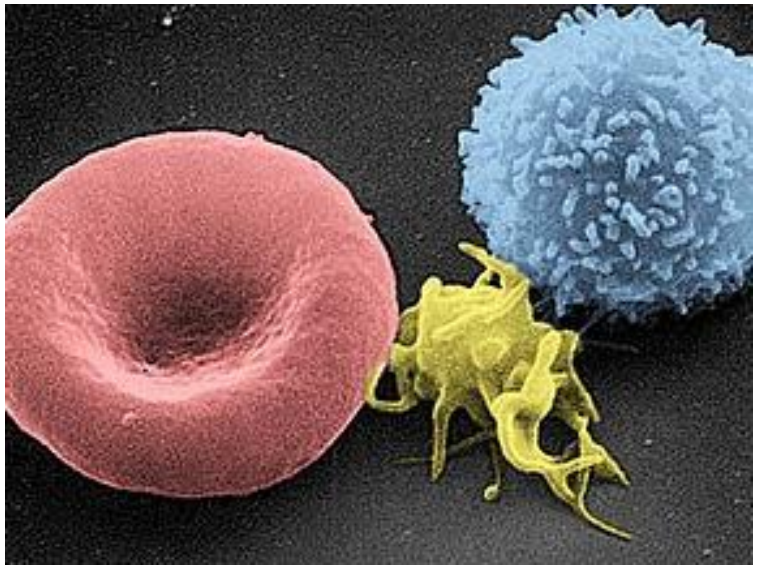
| Class | Statements | By the end of the class, you can... |
|---|---|---|
| 1 | B2.3.1–4 | Explain how a zygote gives rise to specialized cells, what stem cells are, where they are found, and levels of potency |
| 2 | B2.3.5–6 | Compare the sizes of specialized cells, and use surface area-to-volume ratios to explain why cells stay small |

# One cell, two hundred kinds

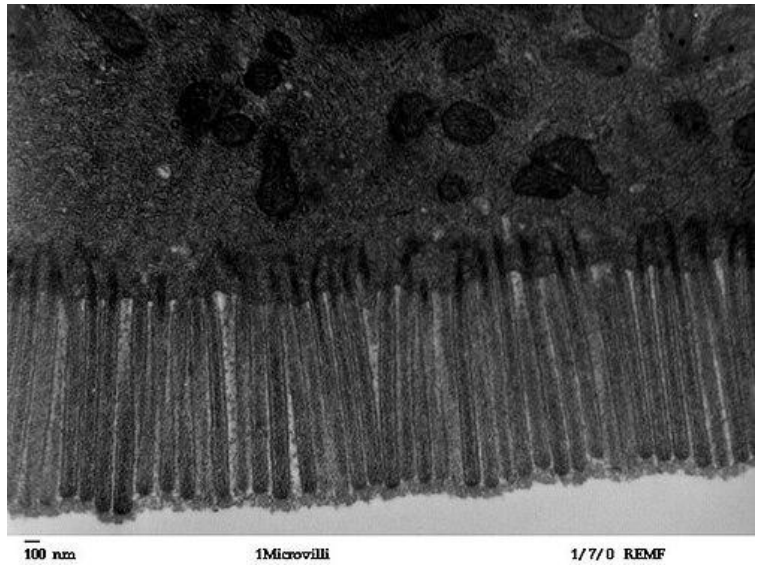
B2.3.1–6



Sperm and egg. Public domain.



Blood cells. NCI, public domain.



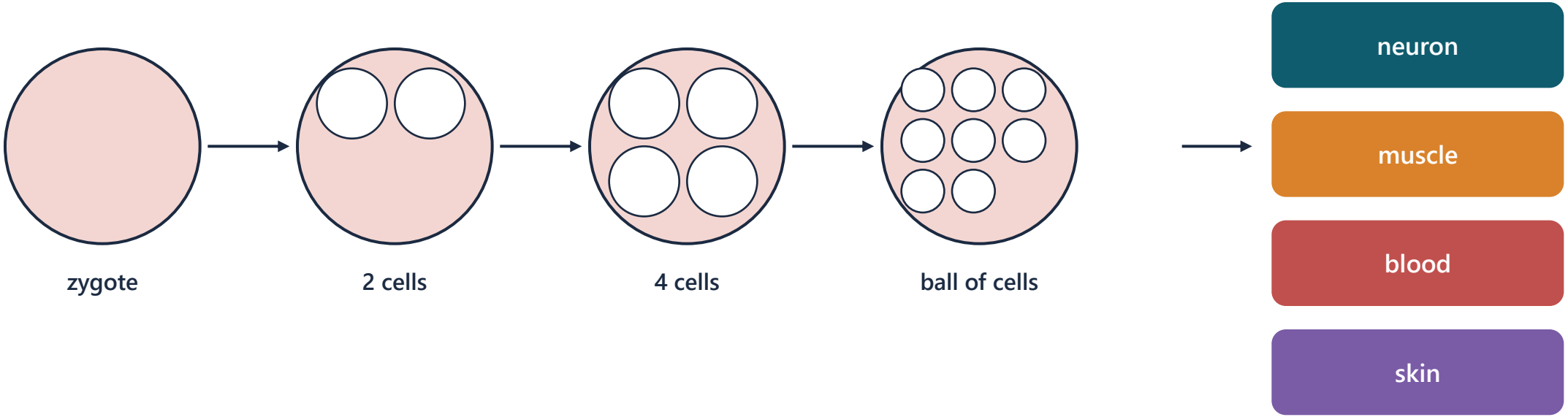

Gut lining cells with microvilli. Public domain.

You began as one cell. Now you have red blood cells, neurons, muscle fibres and gut cells, all with the same DNA but very different shapes and jobs.

This chapter: how cells become specialized, which cells can still become anything, and why almost all cells stay tiny.

# From zygote to specialized cells

B2.3.1

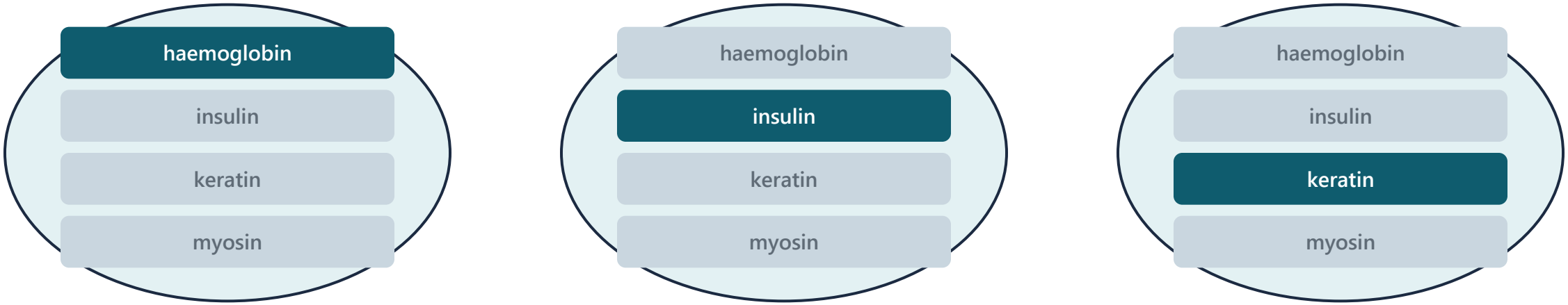



Fertilization makes a **zygote**. It divides by mitosis again and again, making identical, **unspecialized** cells.

Signals then switch different **genes** on in different cells. Each cell follows its own path and **differentiates** into one specialized type.

# Same genes, different cells

B2.3.1

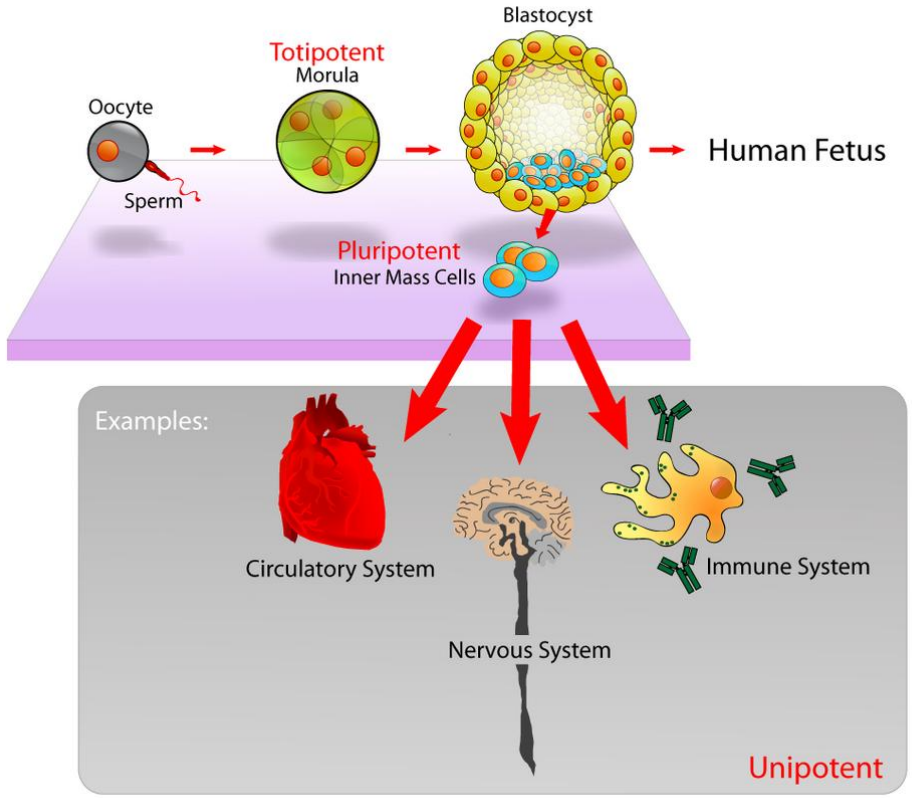



Every cell has all the genes (grey). Each cell type switches **on** only the ones it needs (teal).

So differentiation is about **gene expression**, not about having different genes.

# What makes a stem cell?






Stem cells. Mike Jones, CC BY-SA 2.5.

**Self-renewal** — divides again and again, making more stem cells

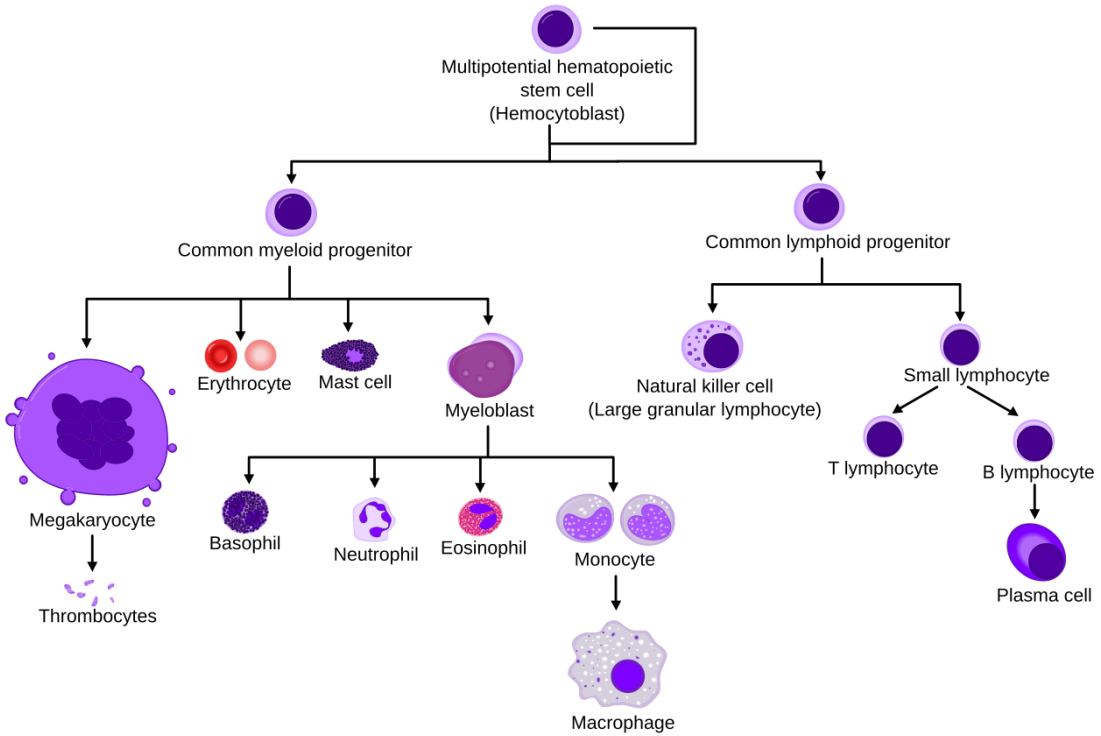
**Differentiation** — can become one or more specialized cell types

## Why they matter

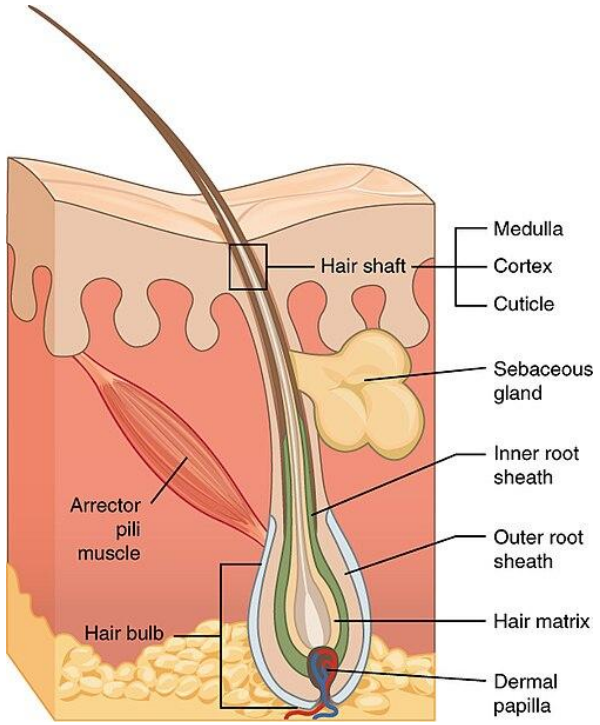
Stem cells grow the embryo, and in adults they replace worn-out cells: skin, blood and gut lining.

# Stem cell niches in adults

B2.3.3




Blood cells from stem cells. CC BY-SA 3.0.




Hair follicle. CC BY 3.0.

A **niche** is a place in the body where stem cells live and are kept ready.

## Two examples

Bone marrow: makes every type of blood cell. Hair follicles: regrow hair and repair skin.

# Three levels of potency

B2.3.4

**Totipotent: any cell type, including the placenta**

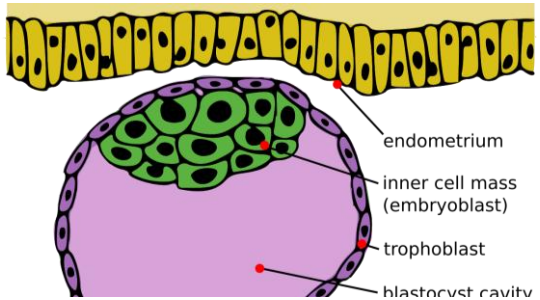
e.g. zygote, very early embryo

**Pluripotent: any body cell type**

e.g. embryonic stem cells

**Multipotent: a few related types**

e.g. blood stem cells in bone marrow




Blastocyst. CC BY-SA 3.0.

As cells specialize, they lose the ability to become other types. Potency goes down step by step.

# Stem cells in medicine

B2.3.2–4

### In use now

Bone marrow transplants give a patient new blood stem cells, for example after chemotherapy for leukaemia.

### Being tested

Growing nerve, heart or insulin-making cells to repair damaged tissue.

### Questions to weigh

Embryonic stem cells can make the most cell types, but taking them destroys an early embryo. Adult stem cells avoid this but make fewer types. Any grown cells must not form tumours.

# Practice: which kind of stem cell?

B2.3.4

| Cells from a 5-day embryo | Cells in bone marrow | The zygote |
| --- | --- | --- |
| can make any body cell but not placenta | make red cells, white cells and platelets | can make a whole organism |

1. Name the level of potency of each.
2. Explain why a skin cell cannot normally become a neuron.

# Working with data: bone marrow transplants

B2.3.3

Survival five years after treatment for one type of leukaemia. These data are invented for practice; the pattern is realistic.

| Treatment | Survival / % |
|---|---|
| **Chemotherapy only** | 35 |
| **Chemotherapy and bone marrow transplant** | 60 |

1. Calculate the increase in survival.
2. Explain how the transplant helps.
3. Suggest why donor and patient tissue must be matched.

# Check-in: class 1

B2.3.1–4

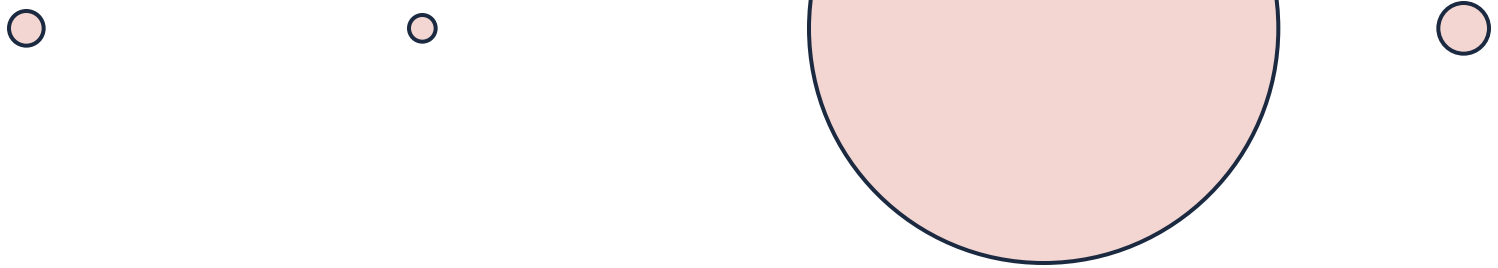
1. Explain how cells in an embryo become different.
2. State two properties of stem cells.
3. Give one stem cell niche in an adult.
4. Distinguish between totipotent, pluripotent and multipotent cells.
5. Give one use and one problem of stem cells in medicine.

**How to use this**

Answer without your notes, then check against slides 4–11.

# Specialized cells come in very different sizes

B2.3.5



**Red blood cell**
8 µm

**Sperm (head)**
5 µm

**Egg**
110 µm

**White blood cell**
12 µm



Hummingbird. CC0.

Some cells are huge (an egg is visible to the eye) and others tiny. A motor neuron can be a metre long but only a few micrometres wide. Size and shape fit the job.

# How size fits the job

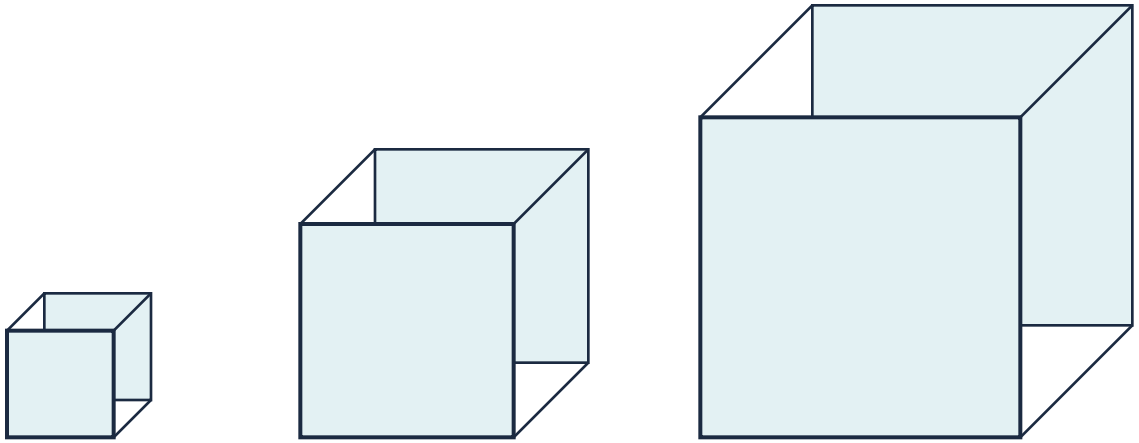
B2.3.5

| Cell | Size | Why |
|---|---|---|
| **Red blood cell** | small | squeezes through capillaries; short distance for $O_2$ |
| **Sperm** | tiny head | little cytoplasm: only needs to deliver DNA, and swim |
| **Egg** | very large | food stores for the early embryo |
| **Motor neuron** | up to 1 m long | reaches from the spinal cord to a muscle |
| **Skeletal muscle fibre** | up to 30 cm | contracts along its whole length |

**Question: why do the egg and sperm, which join to make one cell, differ so much in size?**

# Surface area and volume

B2.3.6



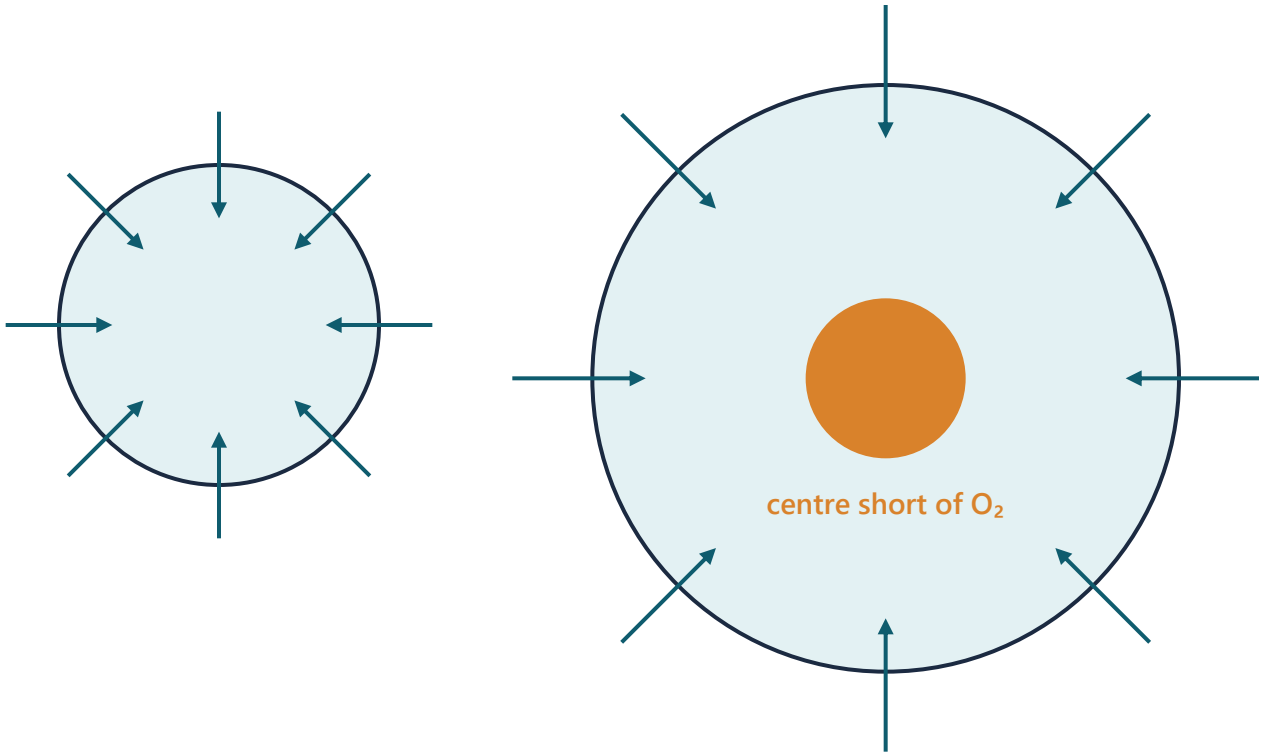

| Side | SA / $cm^2$ | V / $cm^3$ | SA : V |
|---|---|---|---|
| **1** | 6 | 1 | 6 : 1 |
| **2** | 24 | 8 | 3 : 1 |
| **3** | 54 | 27 | 2 : 1 |

As a cell grows, its **volume** (what needs supplies) grows faster than its **surface area** (where supplies get in).

So bigger cells have a **smaller surface area-to-volume ratio**.

# Why cells stay small



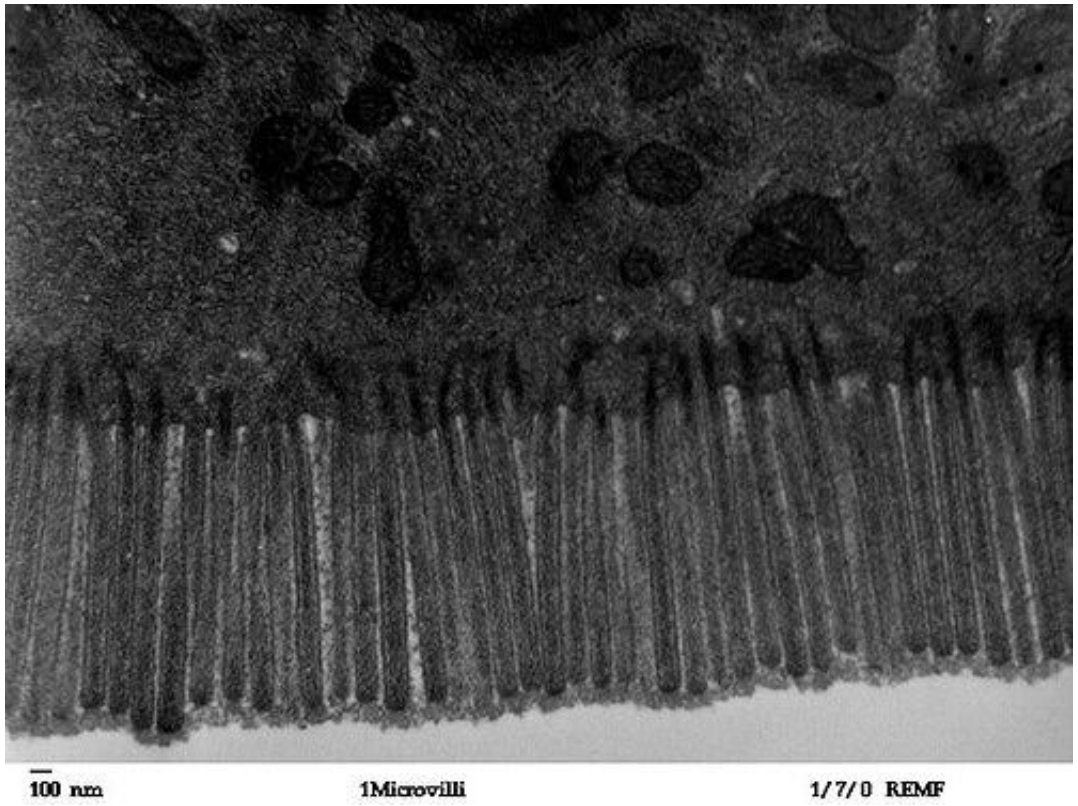



Oxygen and food enter, and waste leaves, through the **surface**. They are used by the whole **volume**.

In a big cell, the centre is far from the surface and the surface is too small to supply it. So cells **divide** before they get too big.

# Getting more surface

B2.3.6




Microvilli on gut cells (TEM). Public domain.

Cells that absorb a lot increase their surface area without growing bigger.

**Microvilli**: finger-like folds of the membrane, as on gut lining cells.

**Flattening**: thin cells, like those lining the alveoli.

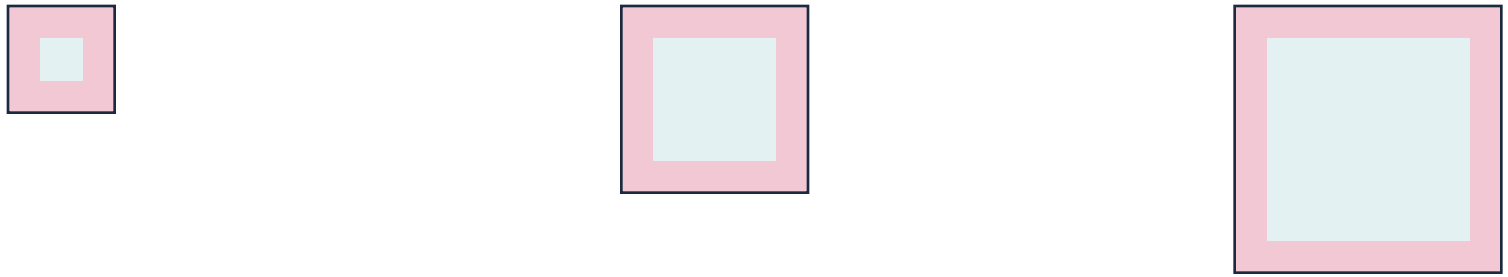
### Link

The same idea applies to whole organs: alveoli, villi, gills.

# Practical: agar cubes

B2.3.6

Cut agar cubes (with indicator) of 1, 2 and 3 cm → Place them in acid → After a set time, cut them in half → Measure the uncoloured centre → Compare with SA : V



Acid diffuses in from the surface. Small cubes change colour all the way through; big cubes keep an uncoloured centre. Safety: wear eye protection with acid.

# Working with data: agar cubes

B2.3.6

Agar cubes in acid for 5 minutes. These results are invented for practice; the pattern matches the real practical.

| Side / cm | SA : V | Coloured / % | Full time / min |
|---|---|---|---|
| **0.5** | 12 : 1 | 100 | 2 |
| **1** | 6 : 1 | 78 | 8 |
| **2** | 3 : 1 | 49 | 30 |
| **3** | 2 : 1 | 35 | 68 |

1. Describe the relationship.
2. Explain the results.
3. Relate them to cell size.

# Practice: calculate SA : V

B2.3.6

1. A cube has sides of 4 cm. Calculate its surface area, volume and SA : V.
2. A cell is a sphere 10 μm across. Another is 20 μm across. Which has the larger SA : V?
3. A flat cell is 40 μm long, 40 μm wide and 2 μm thick. Explain why it can be large but still exchange materials quickly.

# Check-in: class 2

B2.3.5–6

1. Give one very large and one very small human cell, and link each size to its job.
2. Calculate SA : V for a cube with 2 cm sides.
3. Explain why SA : V falls as a cell grows.
4. Explain why cells divide before they get too big.
5. Give two ways cells increase their surface area.

**How to use this**

Answer without your notes, then check against slides 13–20.

# Common mistakes to avoid

| Not this | But this |
| --- | --- |
| **Specialized cells have different genes.** | They have the same genes but express different ones. |
| **Larger cells have a larger SA : V ratio.** | Larger cells have a smaller SA : V ratio. |
| **Adult stem cells can make any cell type.** | Most adult stem cells are multipotent. |
| **All cells are about the same size.** | Sizes range from a few μm to a metre long. |

# Check that you can...

1. **Explain** how differentiation produces specialized cells
2. **Outline** the properties of stem cells and where niches are found
3. **Distinguish** totipotent, pluripotent and multipotent cells
4. **Compare** the sizes of specialized cells and link size to function
5. **Calculate** surface area-to-volume ratios
6. **Explain** why cells stay small

# Exit activity: 5 minutes

**On your own and without notes. Hand this in as you leave.**

1. State two properties of stem cells.
2. Distinguish between pluripotent and multipotent stem cells.
3. Explain why a large cell would struggle to survive.

# Image credits

Sperm-egg.jpg – Unknown author – Public domain

Red White Blood cells.jpg – Electron Microscopy Facility at The National Cancer Institut – Public domain

Human jejunum microvilli 1 – TEM.jpg – Louisa Howard, Katherine Connollly – Public domain

Stem cells diagram.png – Mike Jones – CC BY-SA 2.5

Hematopoiesis simple.svg – A. Rad and Mikael Häggström, M.D. Author info − Reusing imag – CC BY-SA 3.0

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